The International Journal of Oral & Maxillofacial Implants, 06/2025

November 25, 2025

DOI: 10.11607/jomi.11225, PubMed ID (PMID): 40080053 **PAGES 777-789, LANGUAGE: ENGLISH**

Amato, Francesco / Crasto, Gazelle Jean / Spedicato, Giorgio Alfredo / Tarnow, Dennis

Single-Tooth Immediate Implant Placement and Provisionalization in the Esthetic Zone: Infected vs Non-Infected Sites—A 2- to 12-Year Retrospective Clinical Study

Purpose: To investigate and compare the survival rate and the success rate of single-tooth implants placed and restored immediately after extraction in non-infected, acutely infected, and chronically infected sites in the maxillary anterior area. ***Materials and Methods:*** Patients requiring a single-tooth replacement in the maxillary central incisor, lateral incisor, canine, or premolar region were included in the study. Implant sites were divided into three groups based on the presence or absence of an infection: group 1 (control)—non-infected sites (healthy periodontal and endodontic conditions); group 2 (test 1)—acutely infected sites (presence of a periodontal or endodontic or combined abscess and/or fistula); and group 3 (test 2)—chronically infected sites (presence of a periodontal pocket or a periapical lesion with no signs of acute inflammation). The applied protocol was as follows: flapless extraction, thorough debridement of the alveolus, immediate placement of the implant, placement of particulate graft material around the implant, resorbable membrane placement in the facial aspect of the implant for all those cases in which the buccal plate was compromised, and immediate placement of a screw-retained provisional FDP out of occlusion. The following periodontal and endodontic initial conditions were evaluated: gingival recession, probing depth, presence of an abscess and/or a fistula, reason for extraction, presence of a periapical lesion, alveolar buccal wall integrity, distance between the gingival margin (GM) and the alveolar bone crest midbuccally, and implant insertion torque. Implant conditions at the last follow-up included survival rate and marginal bone loss (MBL). Periodontal conditions such as recession, probing depth, and gingival index (GI) and the final esthetic results (pink esthetic score) were also evaluated. Statistical analysis was also performed. ***Results:*** After a mean follow-up of 7 years (range 2 to 12 years), a total of 127 patients were treated and 143 single-tooth implants were placed and immediately restored with a provisional FDP. There were a total of 47 implants in group 1, 56 implants in group 2, and 40 implants in group 3. The survival rate for group 1 was 97.8%, 96.4% for group 2, and 95% for group 3, with no statistical difference between groups (P = .8). A total of five implants failed—one in group 1, two in group 2, and two in group 3—resulting in a cumulative implant survival rate of 96.5%. ***Conclusions:*** The results of this study showed a comparable implant survival rate for single-tooth implants placed and restored immediately in non-infected, acutely infected, and chronically infected sites.

Keywords: acute infection, chronic infection, extraction sockets, immediate placement, infected site, periodontal abscess, single-tooth implant